
Algorithm: Coverage of synthetic images with respect to real images in BiomedCLIP space

Require: Real image sets $\mathcal{R}_{\text{mild}}, \mathcal{R}_{\text{severe}}$

Require: Synthetic sets $\{\mathcal{S}_g\}_{g \in \mathcal{G}}$ (each tagged by severity)

Require: BiomedCLIP encoder f , preprocessor π , thresholds $\mathcal{T} = \{\tau_k\}_{k=1}^K$

Ensure: Table with columns:

- 1: *Generator, Severity, Num_Real, Num_Synth, Threshold,*
- 2: *Num_Real_Covered, Coverage_Fraction, Mean_Min_Dist, Std_Min_Dist*

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3: procedure EMBEDSET( $\mathcal{I}$ )                                ▷ Return unit-normalized embeddings
4:    $\mathbf{Z} \leftarrow []$ 
5:   for each image  $x \in \mathcal{I}$  do
6:      $z \leftarrow f(\pi(x)); \quad z \leftarrow z/\|z\|_2$ 
7:     append  $z$  to  $\mathbf{Z}$ 
8:   end for
9:   return  $\mathbf{Z}$                                            ▷  $\mathbf{Z} \in \mathbb{R}^{|\mathcal{I}| \times D}$ 
10: end procedure

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11: procedure COVERAGE( $\mathbf{R}, \mathbf{S}, \mathcal{T}, g, \text{sev}$ )
12:    $\text{Sim} \leftarrow \mathbf{R}\mathbf{S}^\top$                                 ▷ cosine similarity (unit vectors)
13:    $\mathbf{D} \leftarrow \mathbf{1} - \text{Sim}$                                 ▷ cosine distance
14:    $d_{\min}[i] \leftarrow \min_j \mathbf{D}[i, j]$  for  $i = 1, \dots, |\mathbf{R}|$ 
15:    $\mu \leftarrow \text{mean}(d_{\min}); \quad \sigma \leftarrow \text{std}(d_{\min})$ 
16:   for each  $\tau \in \mathcal{T}$  do
17:      $n_{\text{cov}} \leftarrow |\{i : d_{\min}[i] < \tau\}|$ 
18:      $\text{cov} \leftarrow n_{\text{cov}}/|\mathbf{R}|$ 
19:     record  $\langle g, \text{sev}, |\mathbf{R}|, |\mathbf{S}|, \tau, n_{\text{cov}}, \text{cov}, \mu, \sigma \rangle$ 
20:   end for
21: end procedure

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22:  $\mathbf{R}_{\text{mild}} \leftarrow \text{EMBEDSET}(\mathcal{R}_{\text{mild}})$ 
23:  $\mathbf{R}_{\text{severe}} \leftarrow \text{EMBEDSET}(\mathcal{R}_{\text{severe}})$ 
24: for each generator  $g \in \mathcal{G}$  do
25:    $\mathbf{S}_g \leftarrow \text{EMBEDSET}(\mathcal{S}_g)$ 
26:   if  $g$  tagged as “mild” then
27:     COVERAGE( $\mathbf{R}_{\text{mild}}, \mathbf{S}_g, \mathcal{T}, g, \text{mild}$ )
28:   else
29:     COVERAGE( $\mathbf{R}_{\text{severe}}, \mathbf{S}_g, \mathcal{T}, g, \text{severe}$ )
30:   end if
31: end for
32: return concatenated results table

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Note: Coverage at threshold τ equals the fraction of real images whose nearest synthetic neighbor lies within cosine distance τ in the BiomedCLIP embedding space.
